

Wells, Wires and Dots, and What They Do

KKKC6353 Nanoelectronic Devices · Lecture 2, Part 2 of 3

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What this part covers

Three kinds of trap

Quantum wells, wires and dots, and how they share out the electrons' energies.

Lasers

Colour from the thickness of a layer, and a laser where one electron gives many photons.

Transistors and dots

Fast HEMTs, carbon nanotube transistors, and the quantum dots in screens and imaging.

1

Trapping in one, two or three directions

Name a structure by the directions in which the electron is still free.

Well, wire and dot

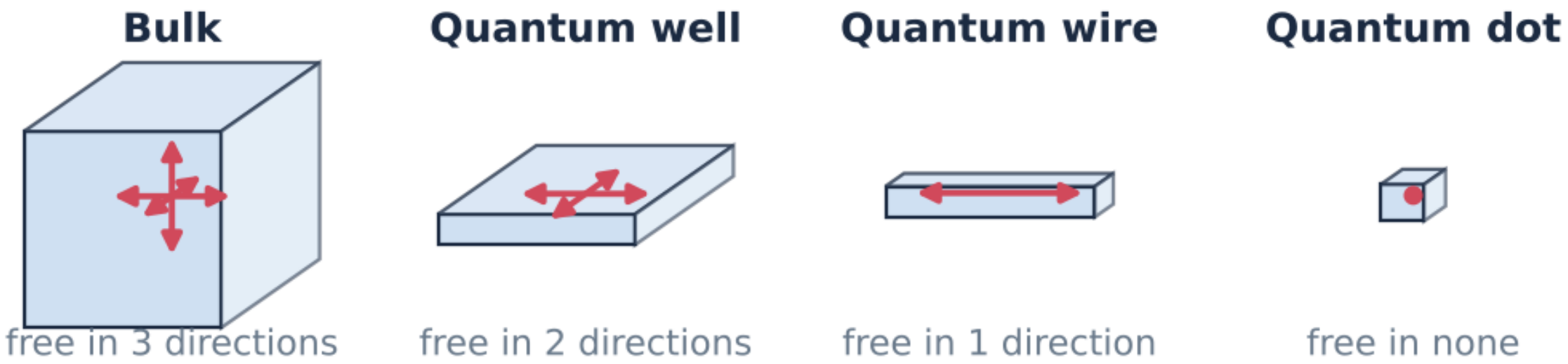


Figure made for this course.

Well: free in a plane, like a puck on an ice rink. **Wire**: free along a line, like a train on a track. **Dot**: not free at all, like a bird in a small cage.

Two things to remember

A dot is 0D, not 1D

The label counts the directions in which the electron is still free: a well is 2D, a wire 1D and a dot 0D.

Walls are made of materials

A thin GaAs layer between AlGaAs layers traps electrons: a heterostructure. The crystals must fit together almost perfectly.

2

How the states are shared out

How many places are there for electrons at each energy?

Places to sit

- Think of a theatre with seats on many levels, one level per energy
- **Density of states** = the number of seats on each level
- **Bulk**: few at the bottom, more higher up
- **Well**: a staircase. **Wire**: sharp peaks
- **Dot**: single sharp levels, like an atom

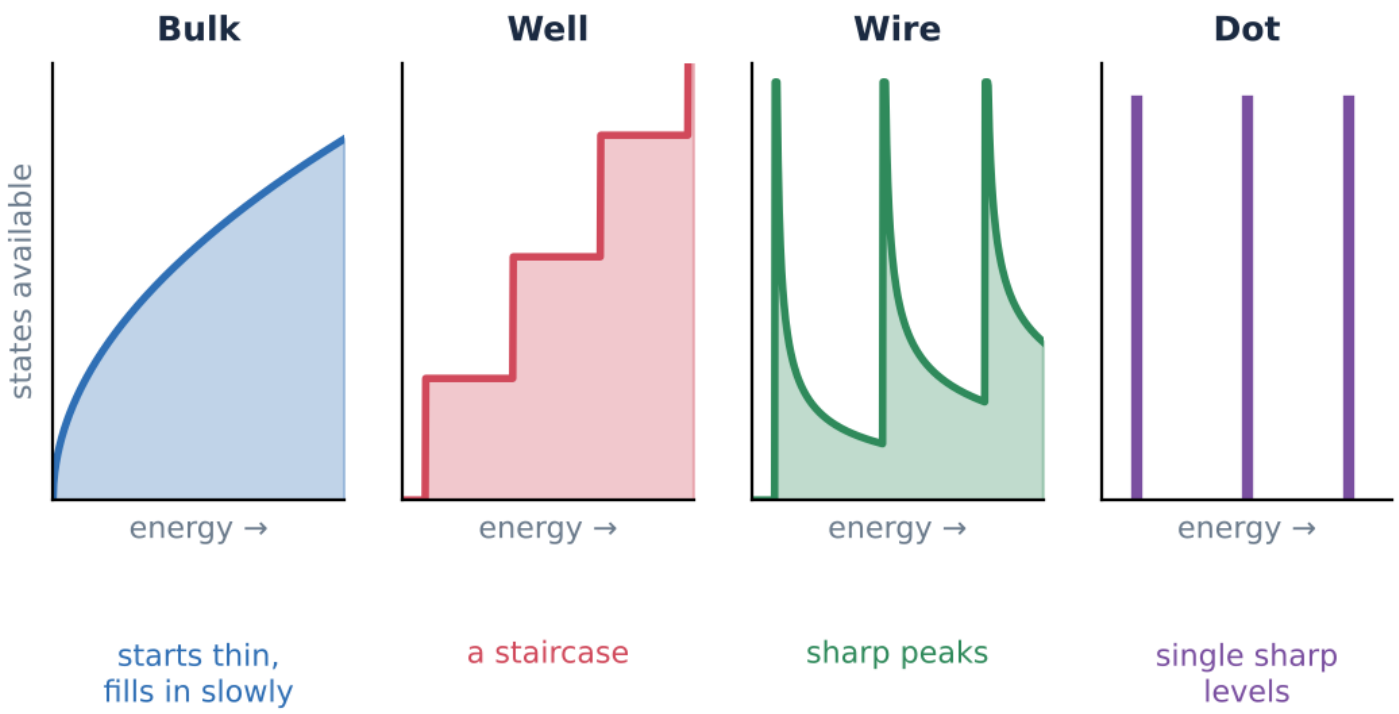


Figure made for this course.

Why it matters: lasers

Bulk

Injected electrons spread over a wide range of energies; many are wasted.

Well

A whole step of places right at the bottom: the laser starts at a much lower current.

Dot

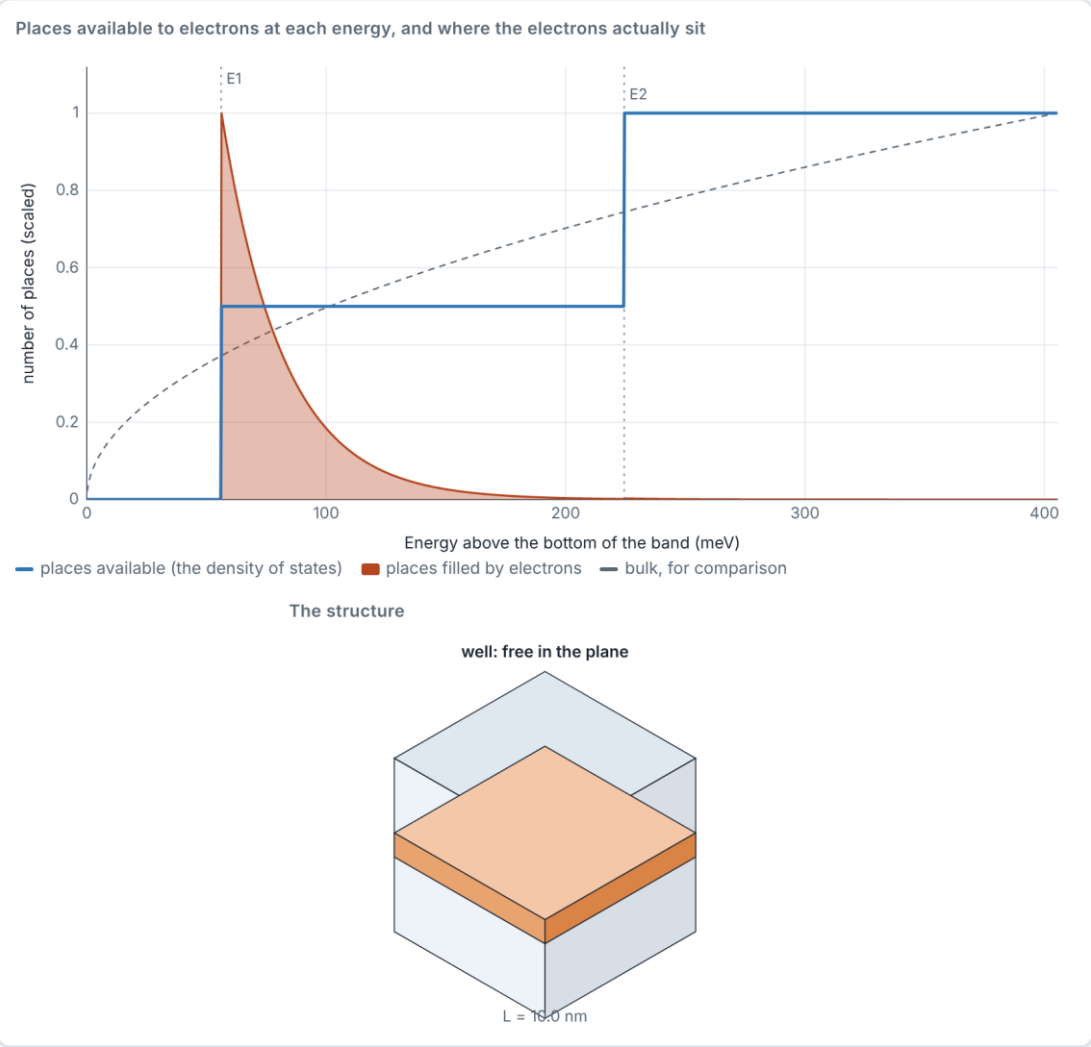
Every electron at one energy, and the laser barely changes as it warms up.

Interactive animation 2

Density of states

faizverse.github.io/nanoelectronic-devices/L2/density-of-states.html

- Start with Bulk: how widely are the electrons spread?
- Switch to Well, Wire, Dot: watch the places bunch up
- Make the trap smaller, then cool it
- When do almost all electrons sit in the lowest level?



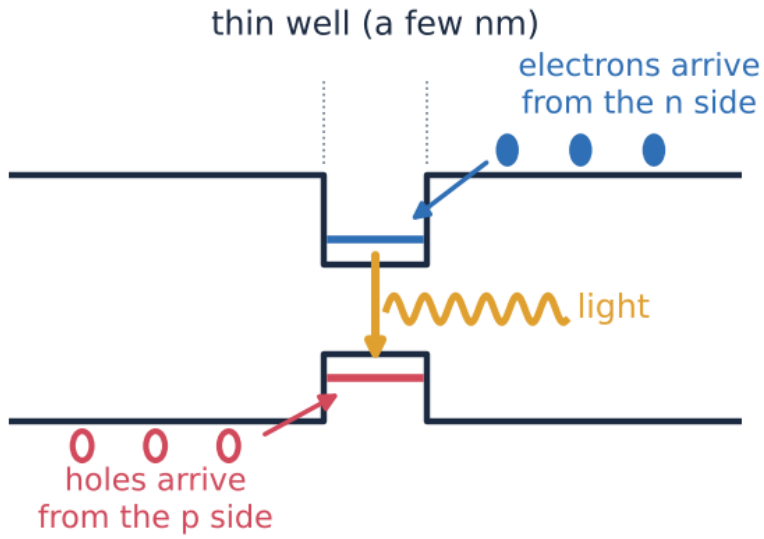
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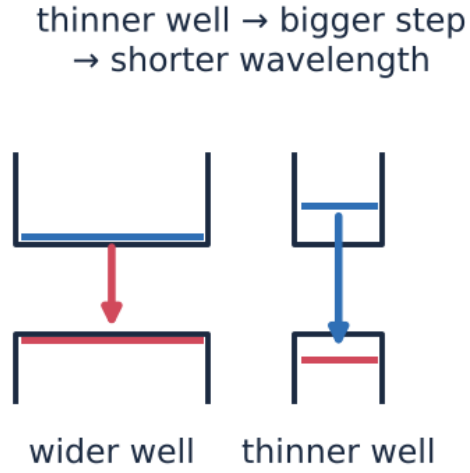
The quantum-well laser

Colour from the thickness of a layer.

Colour from thickness

- Electrons and **holes** fall into a thin well and meet
- Each sits on a **trapped level**, a little away from its band edge
- So the light carries **more energy than the band gap**
- **Thinner well → shorter wavelength**, same material





Drawn schematically. Figure made for this course.

GaAs wells: the numbers

871 nm

Bulk GaAs

848 nm

A 10 nm GaAs well between AlGaAs

816 nm

A 5 nm well: thinner, shorter wavelength

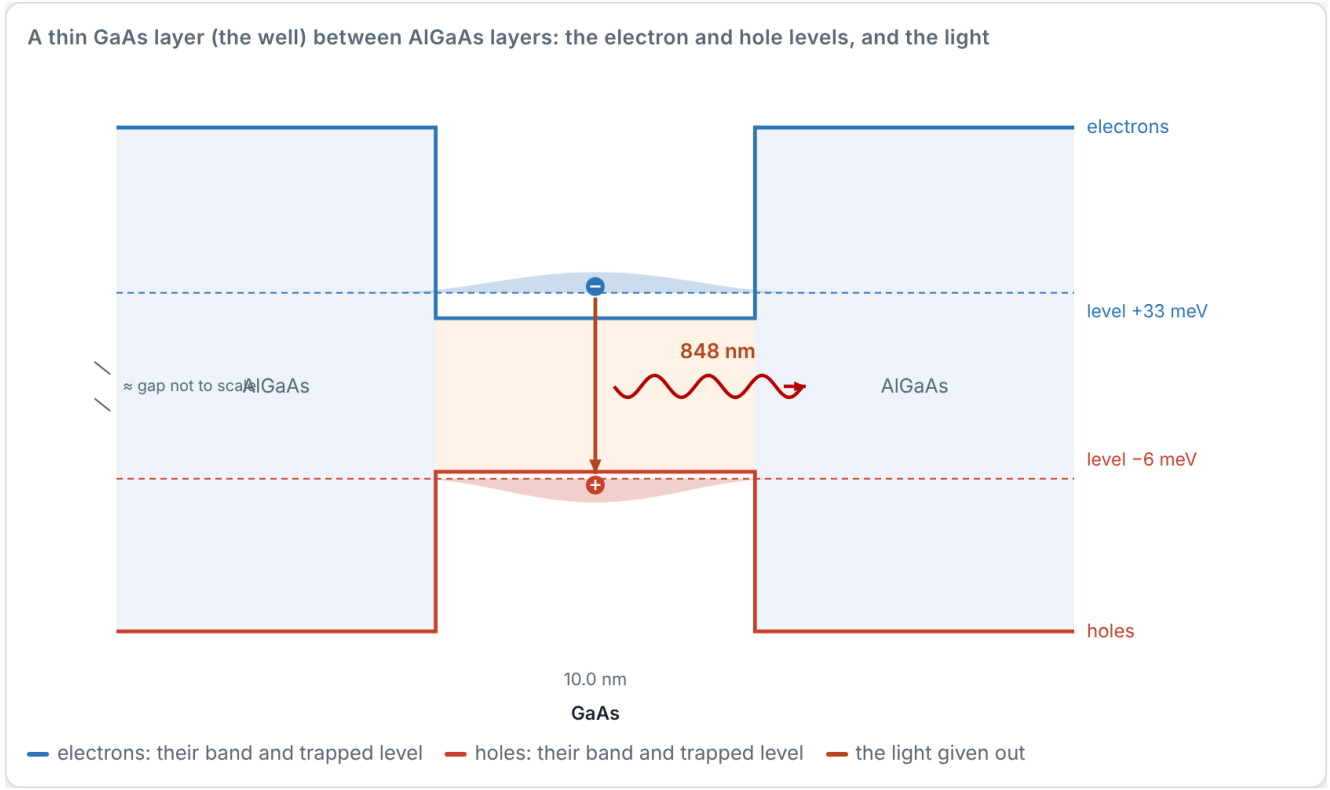
The blue LED behind white LED lighting is a stack of thin InGaN wells (Nobel Prize in Physics 2014: Akasaki, Amano and Nakamura).

Interactive animation 3

Quantum-well emission

faizverse.github.io/nanoelectronic-devices/L2/quantum-well-emission.html

- At 10 nm, compare the wavelength with bulk GaAs (871 nm)
- Predict 5 nm: longer or shorter? Then check
- Which level moves more, electron or hole?
- Find the thickness that gives 830 nm

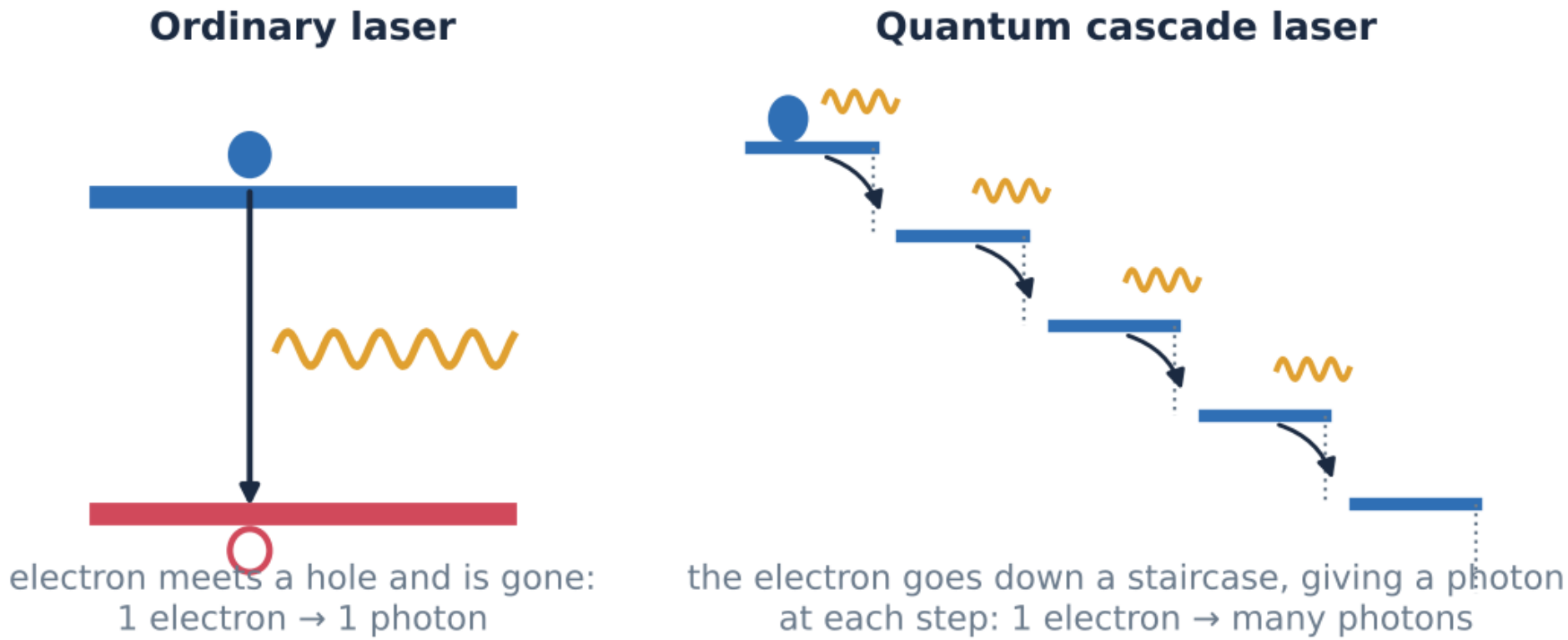


4

The quantum cascade laser

A staircase instead of a cliff.

One electron, many photons



Drawn schematically. Figure made for this course.

The electron drops one step, gives out a photon, **tunnels** into the next step, and drops again, like water down a series of small waterfalls.

Three things follow

1

Only electrons

No holes are needed: the laser is unipolar.

2

Colour from layers

The step size is set by the well thickness, not the band gap.

3

Reaches the infrared

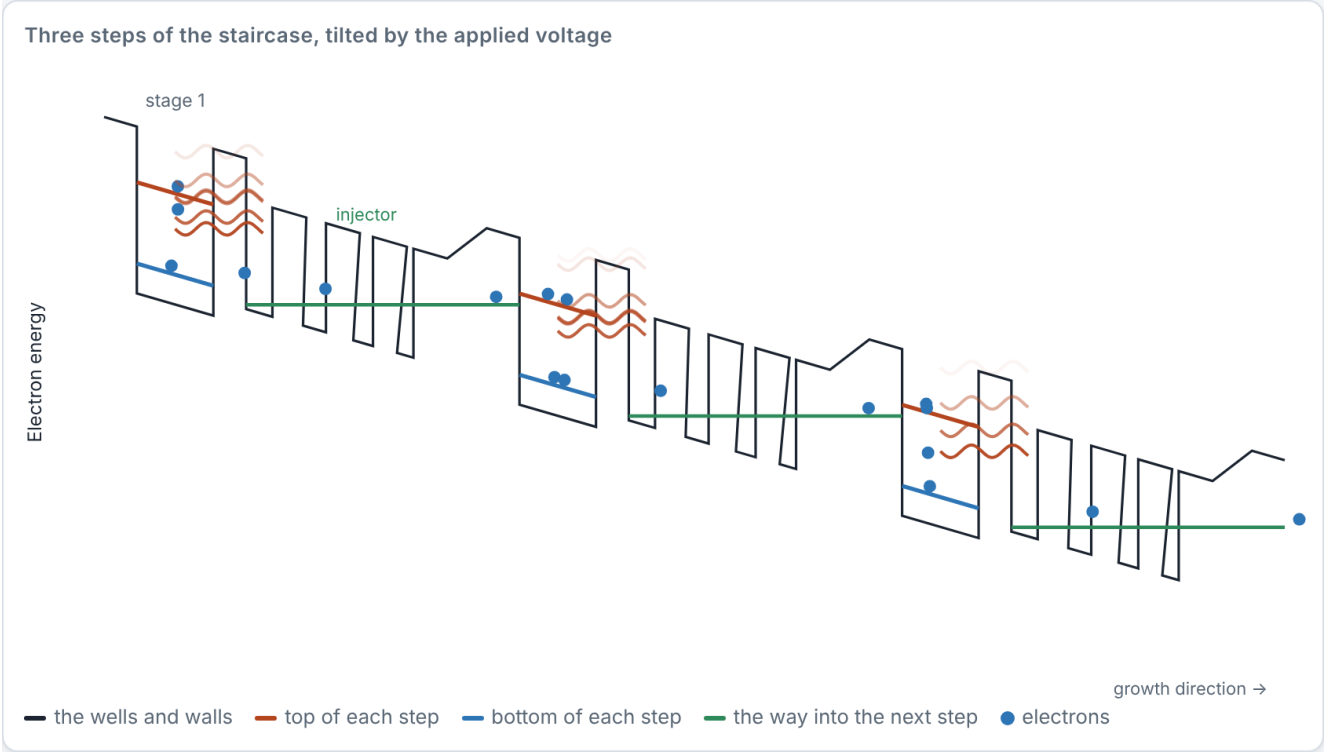
A few to tens of micrometres: sensing tiny traces of gases.

Interactive animation 4

Quantum cascade laser

faizverse.github.io/nanoelectronic-devices/L2/quantum-cascade-laser.html

- Raise the voltage slowly: when does the laser switch on?
- Keep going: what happens when the steps pass each other?
- Widen the active well: longer or shorter wavelength?



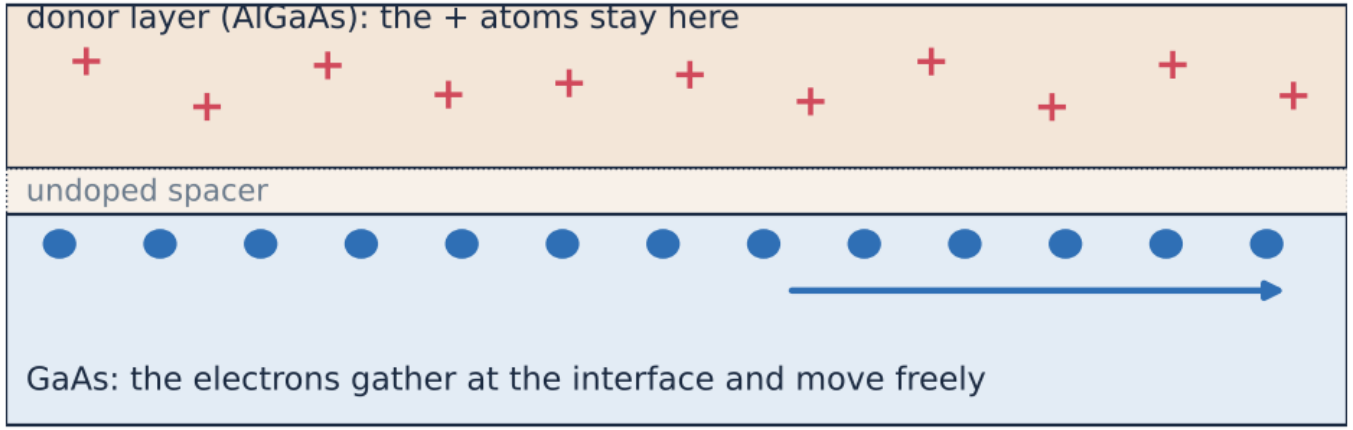
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Two stories from flat and narrow channels

An express lane, and a perfect wire that still has resistance.

The HEMT: an express lane

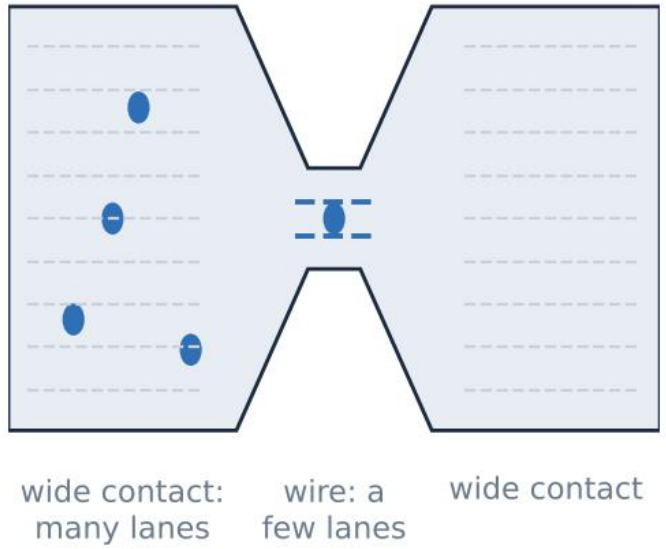
- Donor atoms supply electrons but also **slow them down**
- Put the donors in AlGaAs; the electrons fall into the GaAs
- They travel in a thin sheet **away from the donors**
- **HEMT** (Mimura, Fujitsu, 1980): satellite dishes, base stations
- GaN HEMTs: radar and small laptop chargers

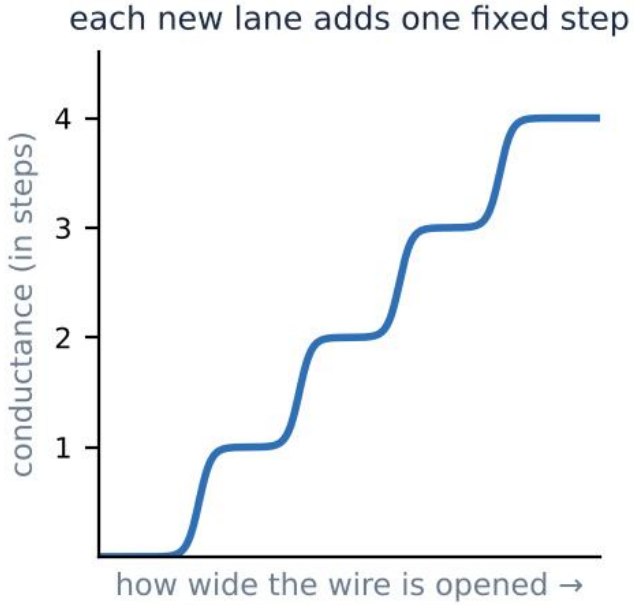


Drawn schematically. Figure made for this course.

A perfect wire still has resistance

- A wire too short to bump into anything: **ballistic**
- Yet the resistance does **not** vanish
- Like a motorway narrowing to a few **lanes**
- Conductance rises in **equal steps**, one per lane
- About **12.9 kΩ** per lane, set by the contacts





Modelled on the 1988 experiments. Figure made for this course.

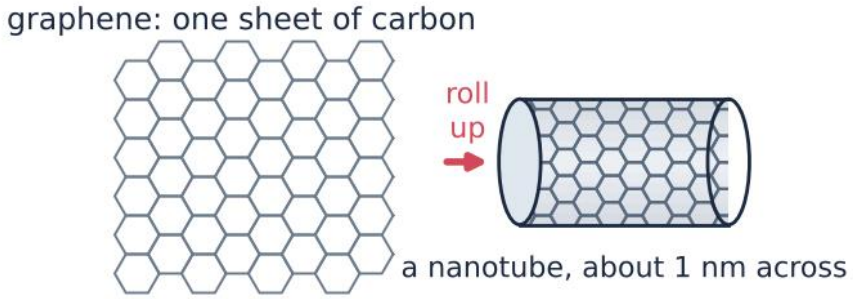
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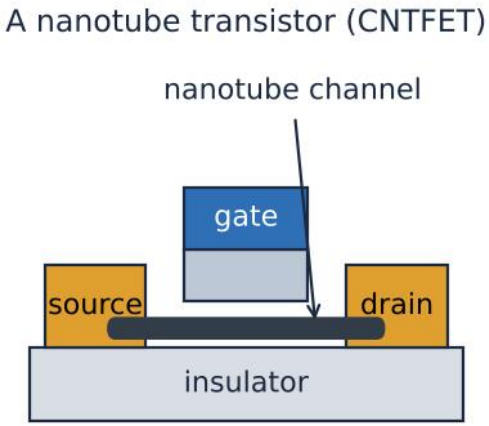
Carbon nanotubes and nanotube transistors

Roll up a sheet of carbon and get a natural quantum wire.

Rolling up graphene

- **Graphene**: one sheet of carbon in hexagons
- Rolled into a tube about **1 nm** across (Iijima, 1991)
- How it is rolled decides **metal or semiconductor**
- About **one tube in three** is metallic
- Thinner tube → bigger band gap





Drawn schematically. Figure made for this course.

What makes nanotubes special

Ballistic

In a short tube electrons cross without bumping into anything.

Fast

Electrons move much more easily than in silicon.

Huge currents

About a thousand times the current density of copper.

Nanotube transistors: milestones

Year	Milestone
1998	First nanotube transistors, from single tubes (Delft and IBM)
2013	First nanotube computer, 178 transistors (Stanford)
2019	A 16-bit microprocessor of 14,702 nanotube transistors (MIT)
2024	A nanotube chip of about 3,000 transistors for neural networks (Peking University)

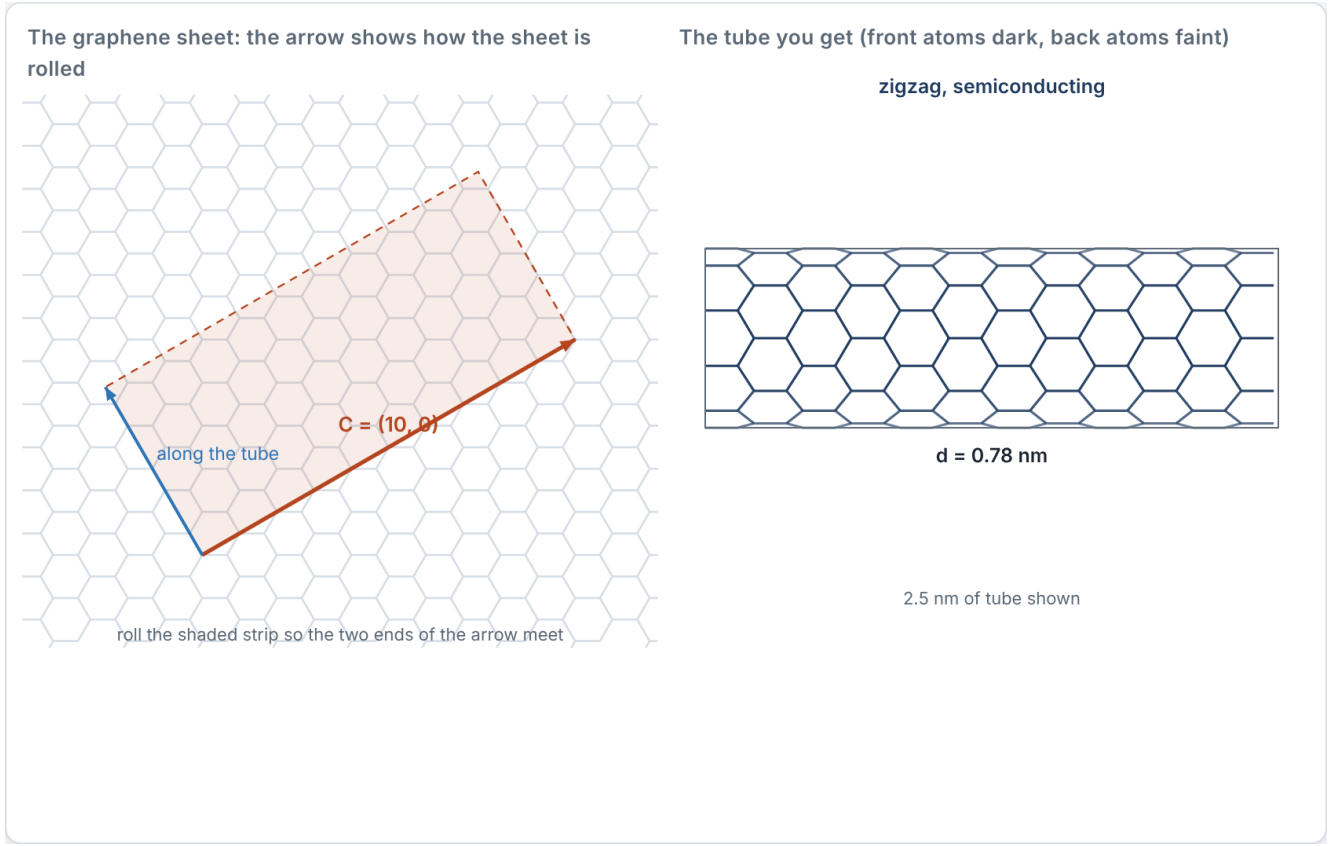
Still hard: every method makes a mix of metallic and semiconducting tubes, and billions must be lined up with good contacts. Lecture 4 returns to them.

Interactive animation 5

Carbon nanotubes: roll your own

faizverse.github.io/nanoelectronic-devices/L2/carbon-nanotubes.html

- (10, 0): predict metal or semiconductor, then check
- (10, 10): why is every armchair tube a metal?
- Press Random ten times: how many could make a transistor?

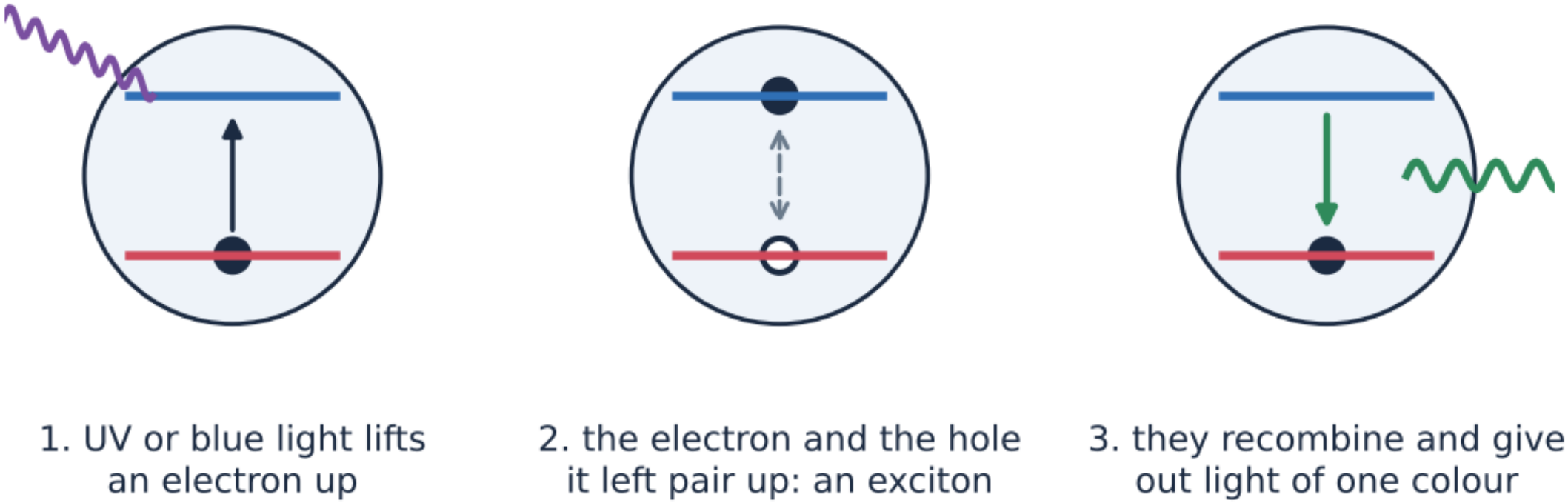


7

Quantum dots

Artificial atoms whose colour you choose.

How a dot gives out light

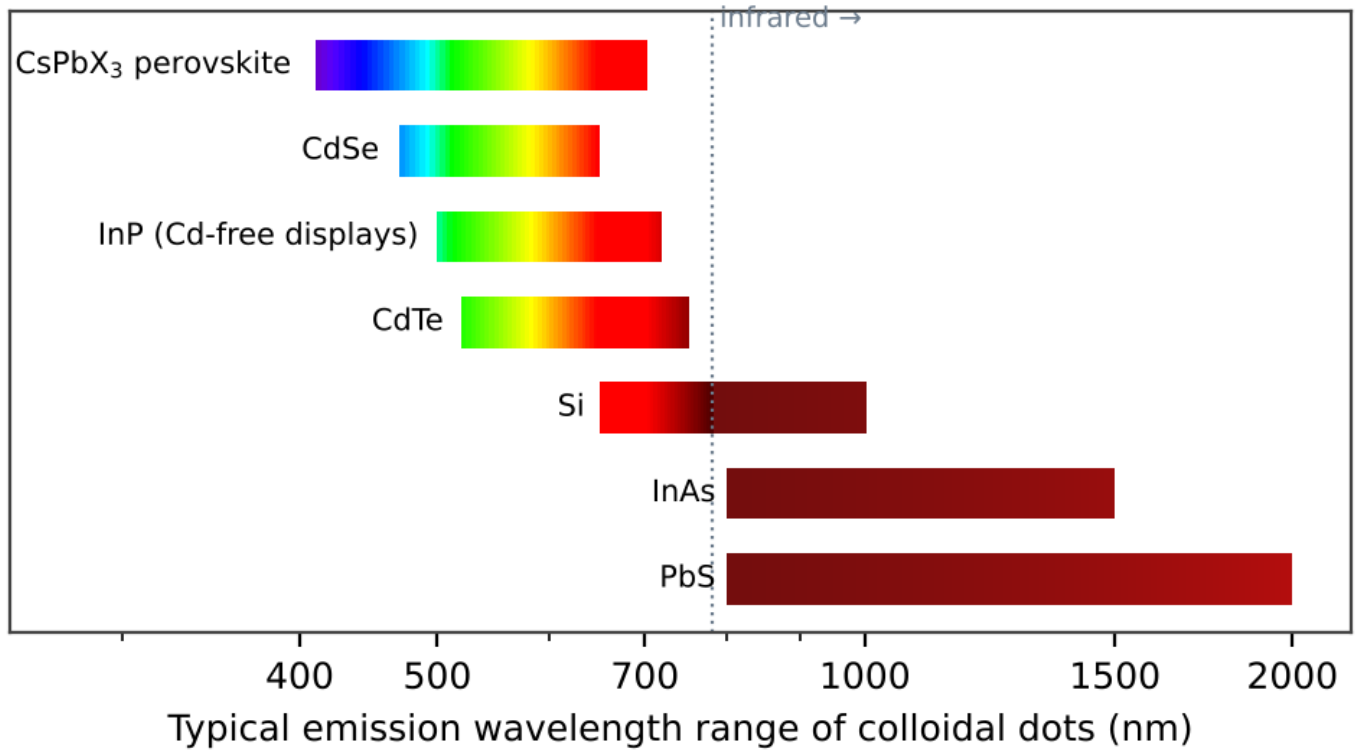


Drawn schematically. Figure made for this course.

Light lifts an electron up; the electron and its hole pair up as an **exciton**; they recombine and give out light of **one pure colour**, set by the size of the dot.

Size and material

- **Small dot → blue**, large dot → red
- But only **within one material**
- Each material's band gap sets its **range**
- 3 nm CdSe glows **green**; 3 nm PbS glows in the **infrared**



Ranges approximate. Figure made for this course.

Dots at work

Screens

Pure red, green and blue: 'QLED' films since 2013, QD-OLED since 2022. Now cadmium-free InP.

Lighting up cells

Bright, slow to fade; several colours tag several targets at once. Mostly in the laboratory for now.

Lasers

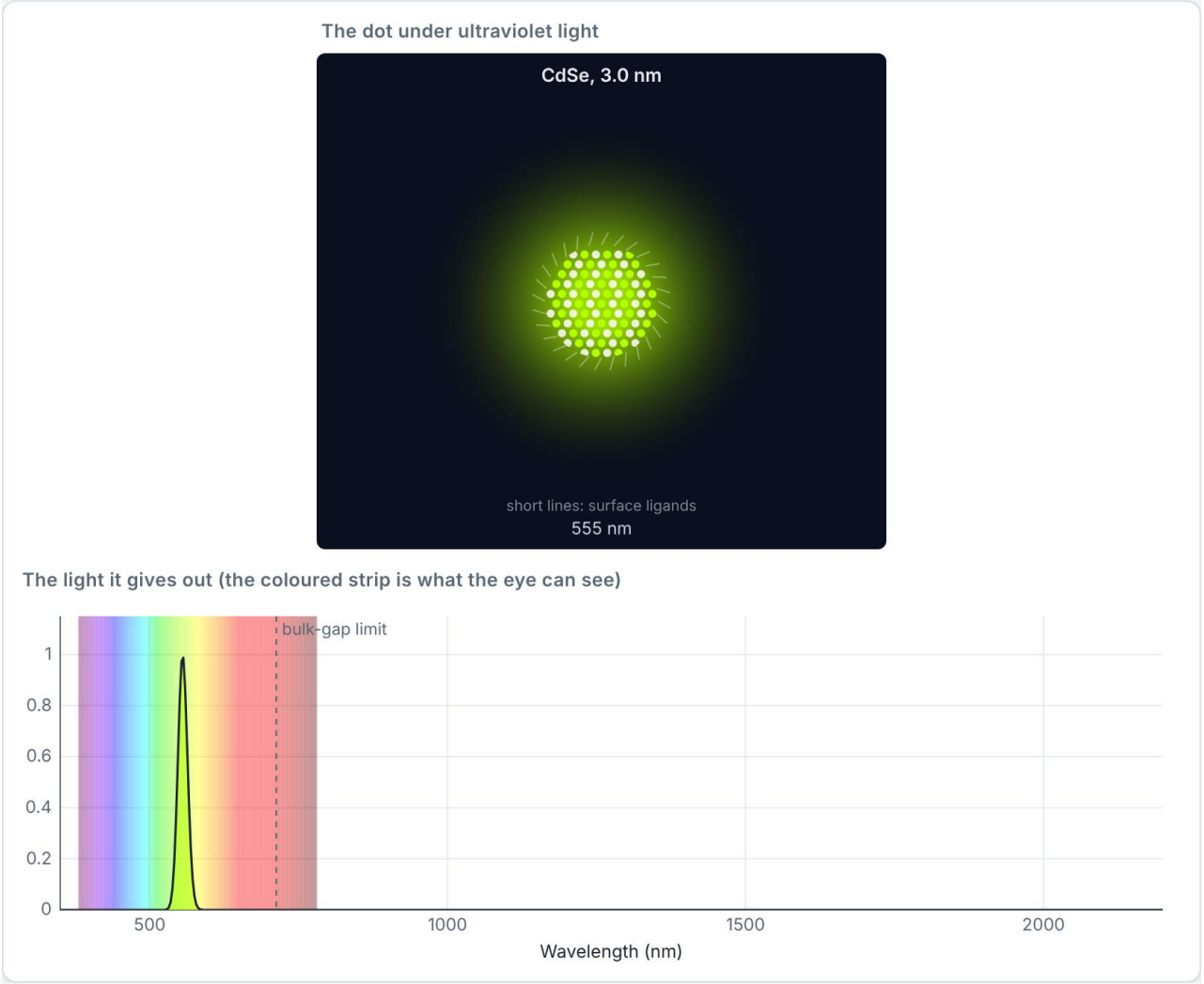
Dot lasers tolerate heat and defects, so they can be grown directly on silicon.

Interactive animation 6

Quantum-dot colour

faizverse.github.io/nanoelectronic-devices/L2/quantum-dot-colour.html

- CdSe: slide from 2 nm to 7 nm and call out the colours
- Which size gives screen green, about 530 nm?
- PbS at 3 nm: where has the light gone?
- Can cadmium-free InP make the same colours?



Part 2 in seven points

1 Wells, wires, dots

Named by free directions: 2D, 1D, 0D.

2 Places to sit

Curve, staircase, peaks, sharp levels.

3 QW lasers

Colour from layer thickness; less current.

4 Cascade lasers

One electron, a photon at each step.

5 Two stories

HEMT express lane; resistance in lanes.

6 Nanotubes

Rolling decides metal or semiconductor.

7 Quantum dots

Colour from size and material.

Credits

Made for this course

All figures were made for KKKC6353 (scripts in figures-source/). Animation screenshots from faizverse.github.io/nanoelectronic-devices.

Sources

Dingle et al. (1978); Mimura et al. (1980); van Wees et al. and Wharam et al. (1988); Faist et al. (1994); Tans et al. (1998); Hills et al. (2019); Brus (1984); Nobel Prize background notes (2014, 2023).